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LETTER

A hybrid statistical–dynamical framework for compound coastal flooding analysis

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Abstract

Compound coastal flooding due to astronomic, atmospheric, oceanographic, and hydrologic drivers poses severe threats to coastal communities. While physics-driven approaches are able to dynamically simulate temporally and spatially varying compound flooding generated by multiple drivers with correlations between some of them, computational burdens limit their capability to explore the full range of conditions that contribute to compound coastal hazards. Data-driven statistical approaches address some of these computational challenges; however, they are also unable to explore all possible forcing combinations due to short observational records, and projections are typically limited to a few locations. This study proposes a hybrid statistical–dynamical framework for compound coastal flooding analysis that integrates a stochastic generator of compound flooding drivers, a hydrodynamic model, and machine learning-based surrogate models. The framework was demonstrated in San Francisco Bay (SF) over the past 500 years with accuracy similar to the physics-driven approach but with much higher computational efficiency. The stochastic generator of compound flooding drivers is developed by coupling a sea surface temperature (SST) reconstruction model with a climate emulator, weather generator, and model of the hydrological and reservoir system. Using reconstructed SSTs as input, the generator of compound flooding drivers is employed to simulate time series of the forcing factors contributing to compound flooding (e.g. surge, waves, river discharge, etc) in SF Bay. A process-based hydrodynamic model is built to predict total water levels varying in time and space throughout SF Bay based on stochastically generated drivers. The machine learning-based surrogate models are then developed from a relatively small library (several hundred) of hydrodynamic model simulations to efficiently predict water levels for compound flooding analysis under the full range of stochastic drivers. This study contributes a hybrid statistical–dynamical framework to better understand the spatial distribution and temporal evolution of compound coastal-fluvial flooding, along with the relative contributions of drivers in complex nearshore, estuarine, and river environments for centennial timescales under past, present, and future climates.

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1. Introduction

Coastal areas are highly vulnerable to compound flooding in which astronomic, atmospheric, oceanographic, and hydrological drivers interact and amplify each other to induce greater impact than when they occur individually [1, 2]. This vulnerability may increase significantly in the future with more frequent and severe compound coastal flooding induced by the increasing frequency and magnitude of various drivers due to climate change [3, 4]. Accurate and efficient analysis of compound coastal flooding would allow a more comprehensive exploration of forcing combinations to provide important information for flooding analysis, which in turn supports the reduction of coastal flood risk and increased resilience of coastal communities [3, 5, 6].

Physics-driven approaches, such as hydrodynamic modeling, have been widely used for compound coastal flooding analyses due to their ability to dynamically simulate complex physical interactions between multiple drivers and accurately calculate spatially varying water levels [3, 7]. Various process-based full-physics models have been developed to simulate flood magnitudes (e.g. TWLs) [8–10], and understand the potentially nonlinear interactions between drivers and their relative importance to extreme compound flooding events [11–13]. Physics-based models are often designed to be two-dimensional (2D), representing the computational domain with meshes and solving the depth-averaged governing equations for 2D flow (Navier–Stokes equations) [14], to strike a balance between computational cost and accuracy [15]. Although accurate, these models are computationally demanding in large and complex coastal areas with multiple drivers [2, 7], especially under stochastic climate and weather conditions [3].

Data-driven statistical approaches have been proposed for an efficient analysis of compound coastal flooding [16, 17] such as joint frequency of key drivers of flooding events [16, 18, 19], and the dependence between drivers [20–22]. Typically, these statistical approaches are developed based on multivariate joint probability analysis. Based on historical observations, reanalysis data, or both, statistical relationships between the parameter of interest (e.g. flood magnitude) and the drivers (as well as between drivers via correlation) are quantified using techniques such as copulas [16, 19, 20]. Data-driven statistical approaches are more efficient in the analysis of compound coastal flooding than physics-driven approaches at a given location provided that the length of the data record is sufficient [17, 22–24]. However, due to limited temporally and/or spatially varying data, statistical approaches are generally unable to explore all possible combinations of forcing conditions and flood scenarios [1, 3, 21]. Their projections are also usually limited to a finite number of locations in space [3, 25] and are not suitable for spatial analysis in this context [26]. In addition, these statistical approaches may not capture the nonlinear and spatially distributed interactions between physical processes [3, 25].

Recently, hybrid statistical–dynamical approaches have been developed for accurate and efficient compound coastal flooding analysis. These hybrid models allow for the simulation of water levels [1, 5, 6, 25], the evaluation of the relative contributions of drivers to extreme compound flooding events [5, 27, 28], and the analysis of the sensitivity of flood metrics (e.g. flood magnitude/TWL) to changing conditions (e.g. climate change) [2, 29]. Most of these approaches combine data-driven statistical approaches and physics-driven approaches. They typically first employ statistical approaches to calculate the joint exceedance return period of compound flood events, and then use computationally expensive physics-driven approaches such as hydrodynamic models to run a limited number of representative flood scenarios from the statistical model. Assuming the extreme flooding event is driven by a specific forcing combination, these hybrid approaches do not necessarily capture the extreme compound flooding event that could be induced by different combinations of drivers that are missing [1, 11]. Also, these approaches are usually not able to provide the temporal (non-stationary) evolution of flooding under a variety of stochastic climate and weather conditions.

Alternative hybrid statistical–dynamical modeling approaches [1, 3, 5] have been developed via the inclusion of a surrogate model which approximates the response of the physics-driven approach (e.g. flood magnitude/TWL) under a range of possible forcing combinations to effectively capture the joint return period of extreme flood events [1, 30, 31]. However, since each coastal area is unique and key compound flooding drivers can vary significantly [12, 22], a site-specific hybrid model is needed for each new location [12, 22].

This study develops a hybrid statistical–dynamical framework for accurate and efficient compound coastal flooding analysis by linking a stochastic generator of compound flooding drivers, a physics-based hydrodynamic model, and machine learning-based surrogate models. The hybrid framework allows for analyzing the magnitude, frequency, and variability of extreme TWLs associated with different return periods, and the relative impact of different drivers to extreme TWLs in SF Bay, California, USA during the past 500 years (June 1501–May 2000). The stochastic generator simulates synthetic time series of joint astronomic, atmospheric, oceanographic, and hydrologic drivers. The generator combines a SST

reconstruction model, a stochastic climate emulator, a stochastic weather generator, and a hydrologic and reservoir system model. Then, a high-fidelity hydrodynamic model is developed to simulate a relatively small number of representative scenarios sampled from the time series of joint processes. Finally, surrogate models are created from the high-fidelity simulations and used to stochastically simulate compound flooding driven by a full range of possible forcing combinations in a computationally efficient manner. The hybrid framework is illustrated by the analysis of extreme TWLs, flood frequency, and an assessment of the relative contributions of the process drivers along the centerline of SF Bay during the past 500 years.

2. Study site

This study illustrates the analysis of compound coastal flooding using a proposed hybrid statistical-dynamical framework in SF Bay, CA, USA, a study site with complex interactions between the nearshore, estuarine and river environments [11]. In addition to SF Bay, the study site (figure 1) also includes the coastal Pacific Ocean and the Sacramento–San Joaquin River Delta to account for multiple water level drivers, such as waves, tides and river discharges (RDs) that contribute to compound flooding in the region [32, 33]. SF Bay has historically experienced coastal flood events that have caused significant infrastructure damage and economic loss and is becoming more vulnerable to compound flooding [34–36]. Much of the research on coastal flooding in SF Bay has focused on physical process-based approaches [11, 32, 33], and the computational burden associated with these approaches limits the possibility of investigating many forcing combinations. Here, a hybrid statistical–dynamical framework is developed to enable a full exploration of all possible forcing combinations under various climate and weather conditions in SF Bay for flooding analysis, which in turn supports coastal flood risk reduction and increased resilience.

3. Methodology

The novel hybrid framework is developed for the first time by integrating a stochastic generator of compound flooding drivers, a physics-based hydrodynamic model, and machine learning-based surrogate models for compound coastal-fluvial flooding analysis with high accuracy and computational efficiency (figure 2). The methodology is briefly presented below and additional details are provided in the supplementary methods and supplementary data sections.

The generator of compound flooding drivers (figure S1) is developed by combining an SST reconstruction model [37], a climate emulator [38], a weather generator [39–41], and a hydrologic and reservoir system model [42, 43] to generate 100 synthetic time series of various atmospheric (sea level pressure-SLP and winds), oceanographic (monthly mean sea level anomaly-MMSL and waves), and hydrological (RD) drivers of compound coastal-fluvial flooding over the past 500 years. The very large number of forcing combinations (100 hourly series, each 500 years long) allow us to estimate extreme compound flood events for different return periods (up to 500 yr) using the proposed hybrid framework. Here, we provide a short summary of each model, with a focus on the novel adaptations to and interconnections between them. Further details on each model can be found in the supplementary material and their original publications. Note that deterministic tides are simulated using UTide [44].

3.1. Sea surface temperature reconstruction

SSTs are reconstructed using a linear regression to estimate annual principal components of SST anomalies over the equatorial Pacific Ocean during the past 500 years from reconstructions of El Niño Southern Oscillation [37]. The variability of errors from this regression are simulated and propagated into an ensemble of SST reconstructions using wavelet autoregression model. The reconstructed time series of SST anomalies serves as input to the climate emulator, TESLA (Time-varying Emulator for Short- and Long-Term analysis) [38].

3.2. TESLA

TESLA [38] is adapted to create multiple traces of the drivers of compound flooding associated with the reconstructed climate of the Pacific Ocean. TESLA uses statistical and machine learning techniques, including copulas and auto-logistic regression, to connect observed climate drivers (SSTs, outgoing long-wave radiation, and SLP) with weather types and oceanographic and atmospheric drivers of extreme water levels. The statistical relationships between weather types and compound flooding drivers (e.g. waves, winds, and SLP) are then used to create hypothetical sequences of hourly flooding drivers given new weather types. Here, using the outputs of the SST reconstruction model, TESLA produces synthetic timeseries of weather types at annual, intraseasonal, and daily scales for the past 500 years. These weather types are used to generate 100 stochastic, hourly flooding drivers (MMSL, waves, winds, and SLP) for the past 500 years based on joint probabilities of weather types and flooding drivers derived from observations of present day climate. The synthetic daily weather type (DWT) time series produced by TESLA also serves as input to a stochastic weather generator.

3.3. Stochastic weather generator

The weather generator is a semi-parametric, multivariate, multi-site model for simulating local

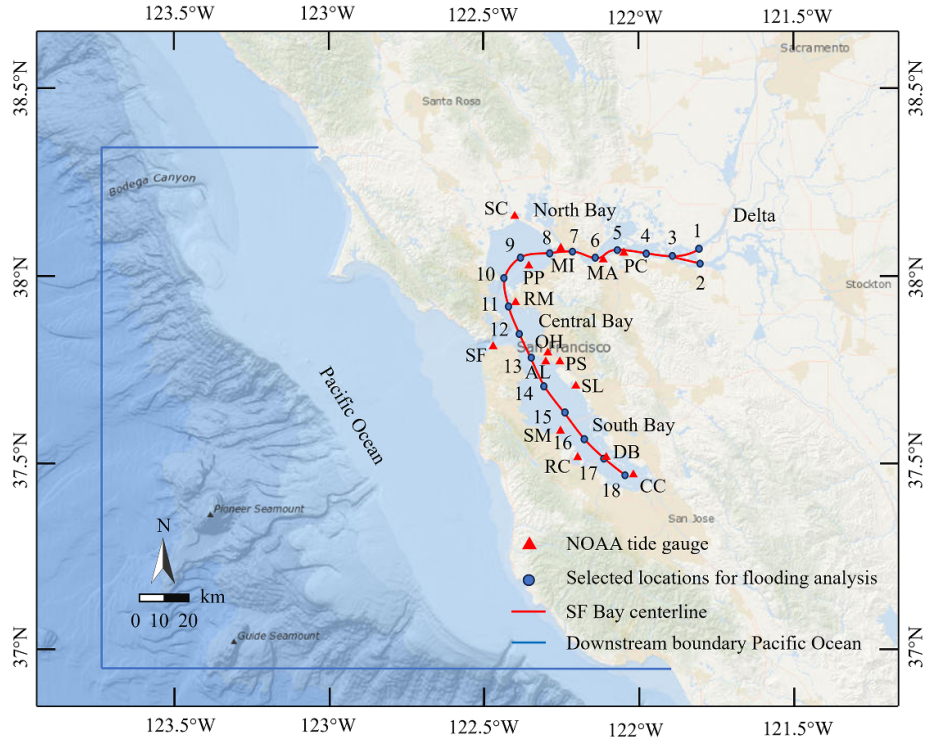

Figure 1. Study site: SF Bay, CA, USA. The red triangles represent NOAA tide gauges. The blue dots denote the 18 locations (coordinations in table S3) along the centerline (red curve) of SF Bay for flooding analysis. SF Bay is classified into North Bay (locations 1–9), Central Bay (locations 10–14), and South Bay (locations 15–18) for convenience of analysis. Seven gauges that are approximately evenly distributed along the centerline of the Bay (i.e. PC, MA, RM, SF, AL, RC, and CC with full names and station IDs in figure S2) are selected for model calibration and validation. Basemap from ArcGIS Pro 3.2.0 (Oregon State University Licensing ArcGIS Pro).

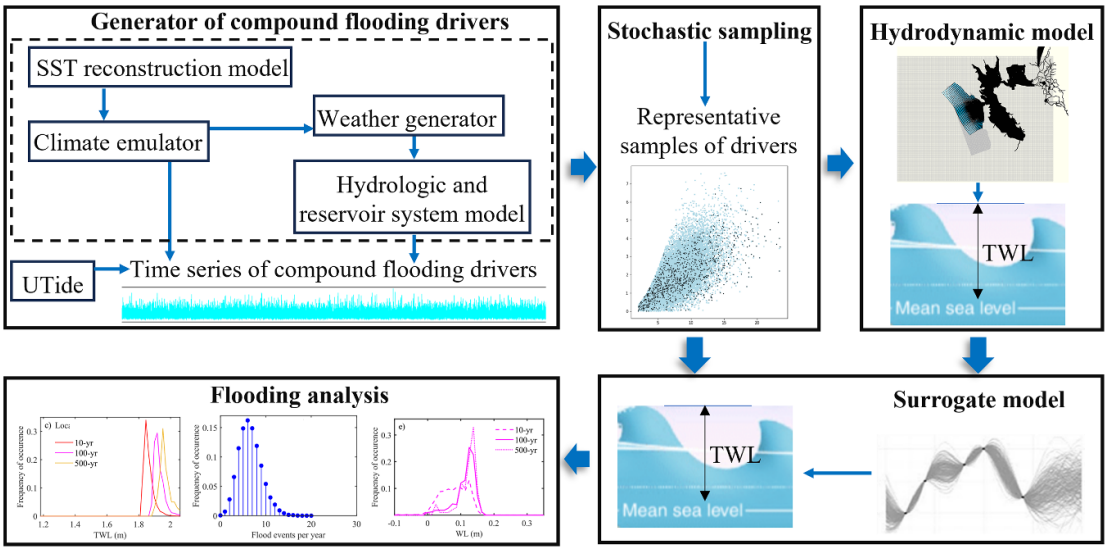

Figure 2. A hybrid statistical-dynamical framework for compound coastal flooding analysis.

weather conditions based on large-scale dynamic and thermodynamic atmospheric mechanisms [39–41]. The model generates sequences of daily local weather (i.e. precipitation-P and temperature-T) using block resampling of historical data with added noise to extreme events, where the blocks are defined by patterns of large-scale atmospheric circulation. We use the 36 DWTs from TELSA to define these blocks, and generate 500 yr long sequences of weather based on the TESLA-simulated DWTs, using daily gridded climate data across all the Sacramento–San Joaquin River basin in California (e.g. daily P (mm), $T_{\min}$

and T_{max} (°C)) [45]. The simulated precipitation and temperature provide the boundary conditions needed for the hydrologic and reservoir system model.

3.4. Hydrologic and reservoir system model

The hydrologic and reservoir system model [42, 43] is a hybrid physics-based and data-driven model developed through combining a rainfall-runoff model and a reservoir system model. The rainfall-runoff model, called Sacramento Soil Moisture Accounting Model (SAC-SMA) [46], is developed for watersheds throughout the Sacramento San–Joaquin Delta [42]. It is built using a number of hydrologic response units (HRUs) determined by climate data (i.e. daily P (mm), T_{min} and T_{max} (°C)) and soil information. The unimpaired flow from HRUs is traced along the river network using the Lohmann routing model [47]. Daily reservoir inflows are simulated using SAC-SMA based on the daily P, T_{min} and T_{max} generated by the weather generator.

A reservoir system model developed for the eight largest reservoirs in the Sacramento–San Joaquin River Basin (SSJRB) [43] is used to estimate outflows from the Delta, considering the impact of reservoir management. The SSJRB model simulates reservoir release policies, tributary gains, and Delta pumping to southern California. Flood control operations are modeled by increasing the water supply release, which is dependent on storage and release. The hydrologic gain into the Delta is determined by the difference between the Delta inflow and the total reservoir outflows. The total daily pumping volume from the Delta is calculated based on estimated water demands in Southern California and constrained by infrastructure capacity and environmental guidelines. Using the daily reservoir inflows provided by SAC-SMA, the SSJRB model creates the daily time series of RDs for Sacramento–San Joaquin Rivers during the 500 years (interpolation into hourly data). Lastly, 100 hourly 500 yr forcing combinations (i.e. tides, MMSL, waves, winds, SLP, and RDs) are collected to generate a relatively small number of representative samples, which define the boundary conditions for the hydrodynamic model.

3.5. Hydrodynamic model

Unique to this study, a Delft3D Flexible Mesh (D3D FM) coupled flow-wave model (figure S2) is developed for SF Bay and used to run hydrodynamic simulations. The boundary conditions in the D3D FM model are defined by the phases of the ten leading tidal constituents, MMSL, wave heights, periods, and directions for swell partitions from Northern and Southern Hemispheres, and local wind sea, winds in East-West and North-South directions (i.e. U, V), SLP, and RDs for Sacramento–San Joaquin Rivers. Maximum Dissimilarity Algorithm [48] is used to generate 750 samples for these 25 variables from the 100 hourly 500 yr forcing combinations. To develop the coupled flow-wave model, the flow model developed in [11] is simplified in terms of grid and boundary conditions (inner black grids), and then coupled with a wave model (outer grey grids) built upon the third-generation Simulating WAves Nearshore computational core [49]. The coupled model is calibrated by adjusting the tidal constituents and bottom friction, and validated for water levels at the seven selected tide gauges along the centerline of the estuary (figure S2). The coupled D3D FM model is used to run 72 h simulations with steady-state boundary conditions to obtain the TWLs in the Bay. Tidal boundary conditions are time-varying during the 72 h simulations to capture nonlinear complexity [5]. The 750 hydrodynamic simulations are used to build quality machine learning-based surrogate models with good and convergent accuracy (e.g. root mean squared error-RMSE) as the training sample size increases.

3.6. Surrogate model of hydrodynamic model outcomes

Surrogate models are developed to efficiently simulate WLs under any forcing combination in any model grid cell based on the 750 D3D FM simulations. We use Gaussian process regression (GPR) for surrogate models because GPR is general, flexible, and capable of capturing uncertainty [3, 5, 50]. An optimized GPR model is built at each of locations 1–18 (figure 1) and the seven selected gauges (figure S2). The GPR models are trained at each gauge/location by automatically optimizing hyperparameters such as covariance (kernel) function using MATLAB's fitrgp function [51] to minimize the five-fold loss of cross-validation [3, 5]. The GPR model is validated at each gauge/location by comparing the predicted TWLs to the TWLs simulated by D3D FM model and/or observed by NOAA. The validation results are presented in figure S3, tables S4 and S5. The uncertainties in the 25 variables are quantified by the GPR model using the joint Gaussian distribution defined by the covariance function, which is further defined by the specific kernel function selected for training the GPR model. Hourly WLs are predicted efficiently by passing any forcing combination into each GPR model.

3.7. Flooding metrics

We adopt three flood metrics including flood magnitude, frequency, and the relative contributions of drivers to extreme flood events for compound flooding analysis considering they can inform flood protection and coastal resilience [1, 2, 11, 52]. All three metrics are quantified based on return level events and/or the corresponding WLs due to each driver.

Return level events are identified by the annual maximum TWLs based on the 100 hourly 500 yr TWLs. For instance, the 5th largest identified after sorting all annual maximum TWLs in a 500 yr time series is the 100 yr event. To calculate the WL due to each driver except tides, we run the GPR model by excluding this driver to estimate the WL due to all other drivers and subtract this WL from the extreme TWL including all drivers. The WL due to tides is directly estimated by running the GPR model excluding all non-tidal drivers [5]. Flood magnitude is quantified by the return level TWLs. The number of extreme water level events above certain flood thresholds per year is counted. Flood frequency of each given flood threshold is also calculated based on the exceedance probability of annual maximum TWLs in each simulation. Here we consider four flood levels (i.e. nuisance, minor, moderate, and major flooding [53, 54]; see the supplementary material for details about flood thresholds). At each location, the WL due to an individual driver is divided by the extreme TWL to obtain the relative contribution of each driver. The non-linear interactions between forcing drivers are not taken into account in the calculation of the relative contribution due to the difficulty in accurately calculating these small magnitude values using the stochastic (GPR) models. All reported WLs are relative to mean sea level (MSL).

4. Results

We present the results of compound flooding analysis along the centerline of SF Bay (18 representative locations) with the focus on flood magnitude, frequency, and variability of extreme WLs, and the relative contribution of different drivers to extreme TWLs. All the probability distribution functions in this study are estimated using the kernel density estimator.

4.1. Extreme TWLs

The distributions of all the extreme TWLs that are equivalent to 10 yr, 100 yr, and 500 yr events are calculated across the 100 simulations at each location. Figures 3(a)–(c) show the probability distribution of 10 yr, 100 yr, and 500 yr extreme TWLs at locations 5, 12, and 18. Note that we selected these three locations, with one each in the North Bay (5), Central Bay (12), and South Bay (18), as representative locations to illustrate the spatial flooding analysis in SF Bay using the proposed hybrid framework. The extreme TWLs computed in different return periods show a clear variability. Extreme TWLs in longer return periods peak at larger values and have a larger variance.

The mean extreme TWL is calculated by averaging all extreme TWLs (over the 100 simulations) in a given return period at each location. Figures 3(d) and (e) present the spatial distribution of the mean extreme TWLs in 1 yr, 5 yr, 10 yr, 50 yr, 100 yr, and 500 yr period. The mean extreme TWLs vary spatially along the centerline of SF Bay (figure 3(e)) and near the Delta mouth (figure 3(d)). Overall, the extreme TWL increases from north to south in all return periods and the largest difference can be as high as 0.77 m between locations 18 and 3 in a 1 yr event. The variability of the TWLs between different return periods towards the Delta is larger than in South Bay. For instance, the extreme TWL associated with the 500 yr event is 0.43 m and 0.35 m higher than the 1 yr event at locations 3 and 18 respectively. The difference between extreme TWLs is even larger near the Delta mouth, i.e. 0.62 m at both locations 1 and 2. Note that the extreme TWL near the Delta mouth (locations 1 and 2) is larger than that where two rivers converge (location 3) for the 500 yr event while smaller for other return-level events.

4.2. Flood frequency

The number of extreme water level events in which nuisance, minor, moderate, and major flood thresholds are exceeded per year is calculated for each simulation. Figures 4(a)–(c) present the distribution of nuisance flood events per year at locations 5, 12, and 18. The probability distribution of nuisance flood events per year shows different variability over location. Nuisance flooding at location 5 is more likely to experience around four events per year, but it might never happen (i.e. zero events) or occur as many as 31 events each year. However, nuisance flooding at locations 12 and 18 almost occur every year, and are most likely to occur around nine and six times each year, respectively. There could be as many as thirty nuisance flood events at location 12 while only as many as twenty nuisance flood events at location 18 each year.

The mean flood frequency for each of the four flood thresholds at each location is estimated by averaging all the flood frequencies over 100 simulations. The spatial distributions of flood frequency for the four flood levels are shown in figures 4(d) and (e). The flood frequency, except for major flooding, has spatial variability in SF Bay. Overall, the flood frequency increases slightly from North Bay to South Bay. Also, the flood frequency shows a similar trend to the extreme TWL in terms of spatial variability (figure 3(e)). Near the Delta mouth (figure 4(d)), locations 1 and 2 have higher minor and moderate flood frequencies but lower nuisance frequency than location 3.

4.3. Relative contributions of drivers to extreme TWLs

The relative contribution, and associated variability (i.e. the variability in the WL due to each driver), of each driver to extreme TWLs with different return periods are calculated (figure 5 at location 5 in the North Bay). The relative contribution of each driver to extreme TWLs display a clear variability. The WLs associated with each driver are less variable for longer

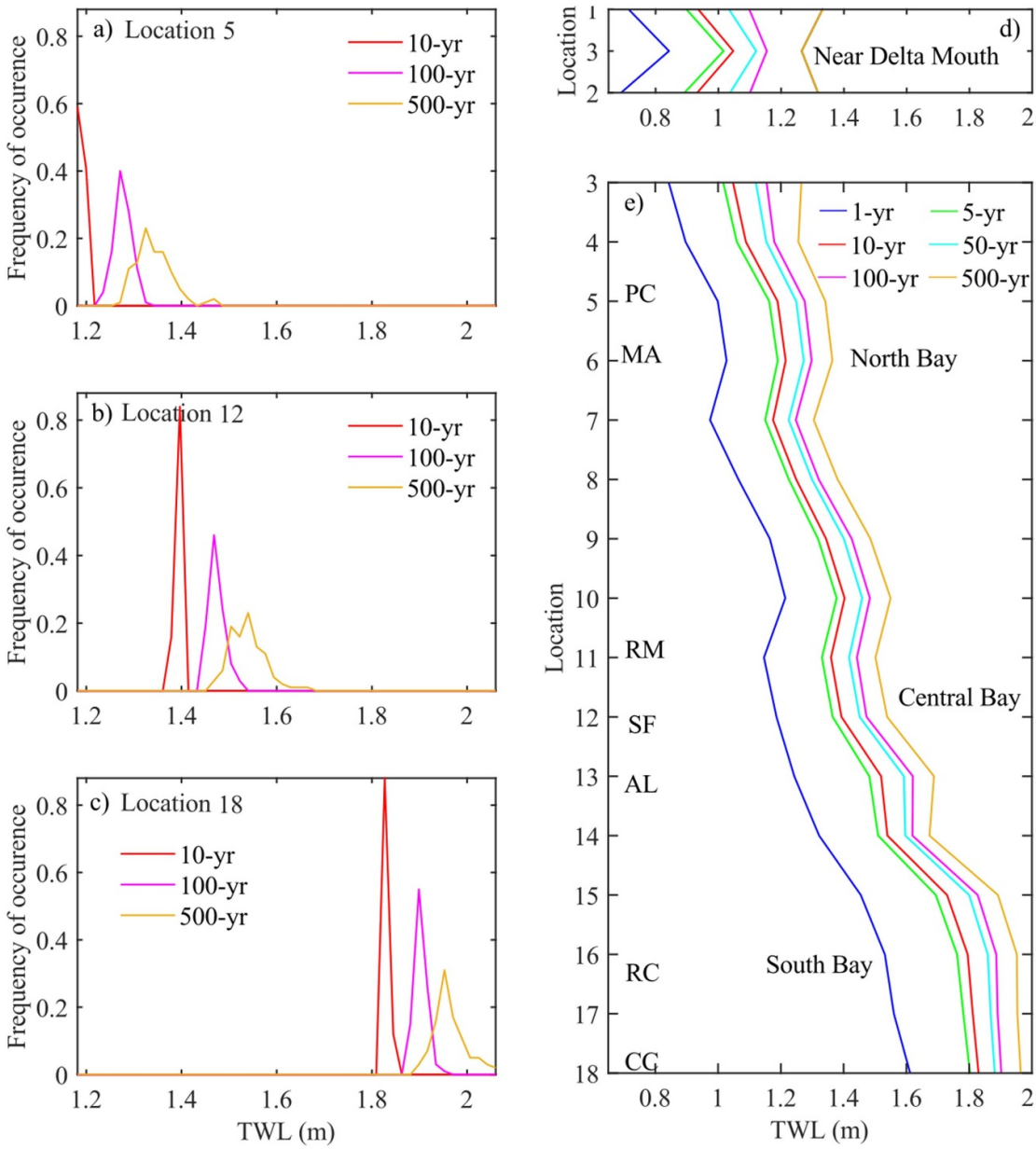

Figure 3. (a)–(c) Probability distribution of extreme TWLs at locations 5, 12, and 18 (close to one NOAA gauge in North Bay, Central Bay, and South Bay, respectively). (d), (e) Spatial distribution of mean extreme TWLs near the Delta mouth and along the centerline of SF Bay (note location 3 is shared by both subregions).

return periods (i.e. smaller range and more peaked distribution). As return period increases, all the contributions due to non-tidal drivers shift to a larger magnitude (i.e. peaking at around 0.1 m) while the contribution due to tides becomes smaller.

The average relative contributions of each driver to different return level TWLs across the 100 simulations are calculated at each location (figure 6). The relative contributions to extreme TWLs show a clear spatial variability. Overall, tides dominate the extreme TWLs and their contribution increases from north to south. Non-tidal drivers make a smaller contribution that increases from south to north, especially the contribution of RD. As return period increases, however, the contribution of the tide decreases while the contribution of the non-tidal drivers increase, especially moving from the 10 yr to the 100 yr return level events. Near the Delta mouth (figures 6(a)–(c)), the contribution of the non-tidal drivers is larger than the contribution of tides for the 500 yr return level event.

The relative contributions of each non-tidal driver (MMSL, waves, SLP, wind, and RD) to the non-tidal component of extreme TWLs can also be explored with the stochastic simulated time series produced by our hybrid approach (figure 7). As expected, the RDs contribute the most to the non-tidal water levels near the Delta (especially near the Delta mouth in figures 7(a)–(c)). However, the RDs' contribution decreases quickly over distance from near the Delta mouth to around location 8. As the extreme return period increases, RDs contribute

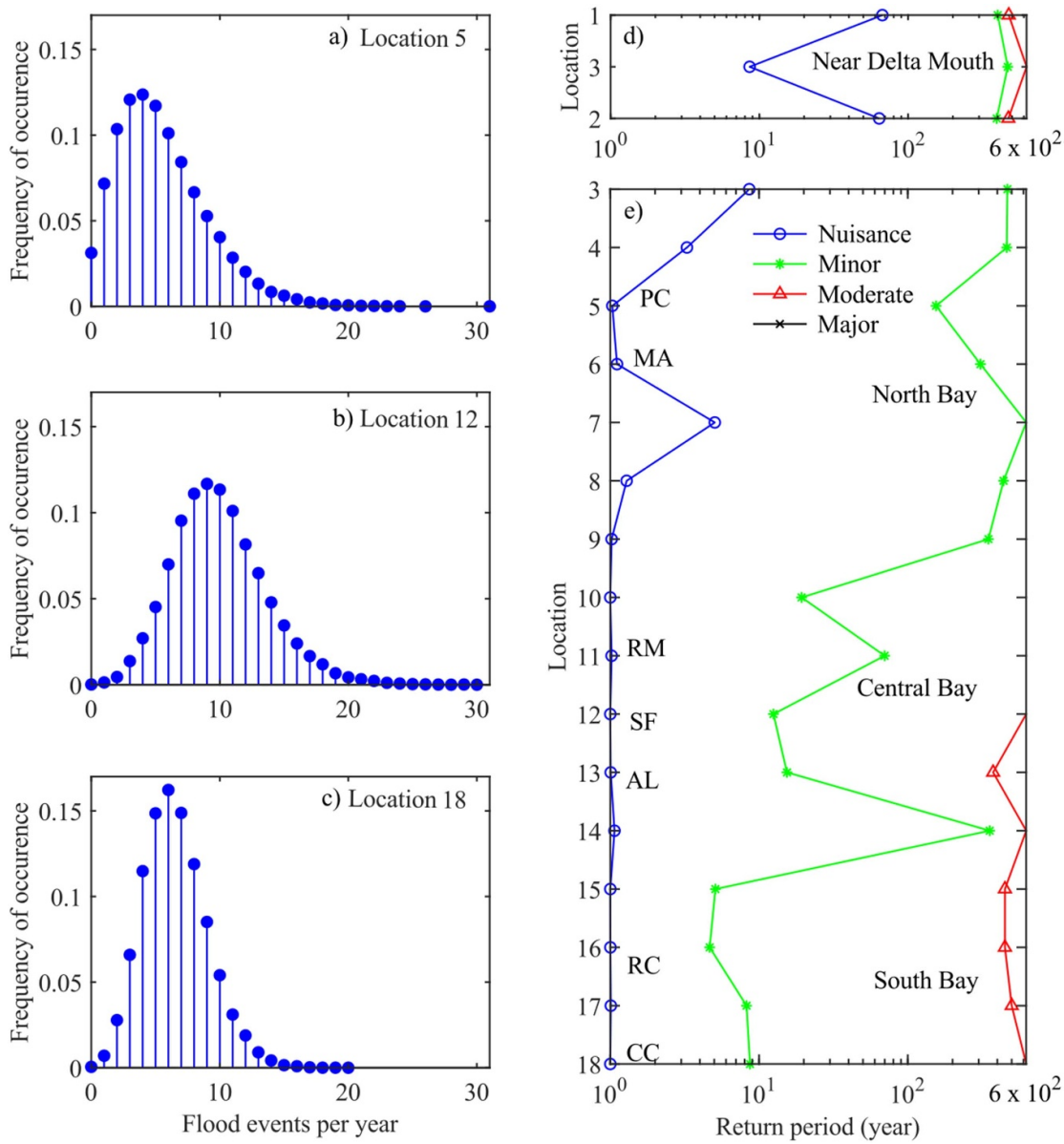

Figure 4. (a)–(c) Probability distribution of nuisance flood events per year at locations 5, 12, and 18. (d), (e) Spatial distribution of nuisance, minor, moderate, and major flood frequencies near the Delta mouth and along the centerline of SF Bay. Note that moderate flood frequencies at some locations and major flood frequencies at all 18 locations are not plotted if no such flood events are found in analysis.

relatively more than the other drivers to the non-tidal water level near the Delta. South of approximately location 8 (including locations 8–9 in North Bay, Central Bay, and South Bay), the relative contribution of each driver to the non-tidal component of extreme TWLs does not vary with return period. Overall, the non-tidal drivers other than RDs have approximately the same relative contribution at locations 8 through 18.

5. Discussion

The compound flooding analysis afforded by our proposed hybrid approach, reveals clear variability in the flood magnitude, frequency, and relative contributions of different drivers to extreme water levels at a particular location and across locations in different return periods, which provides important information for compound coastal flood management in SF Bay. The much larger TWL in South Bay does not suggest more severe flooding, as it is simply a function of tidal amplification over distance [11]. The much greater increase in extreme TWL in a longer return period near the Delta than at other locations is due to the influence of RD [11]. The varying extreme TWLs are the basis of additional flood analysis in terms of flood frequency and relative contributions of drivers to extreme flood events. The proposed hybrid framework enables us to identify these extreme TWLs by exploring a large number of forcing combinations, which would be challenging using the dynamical approach or statistical approach alone.

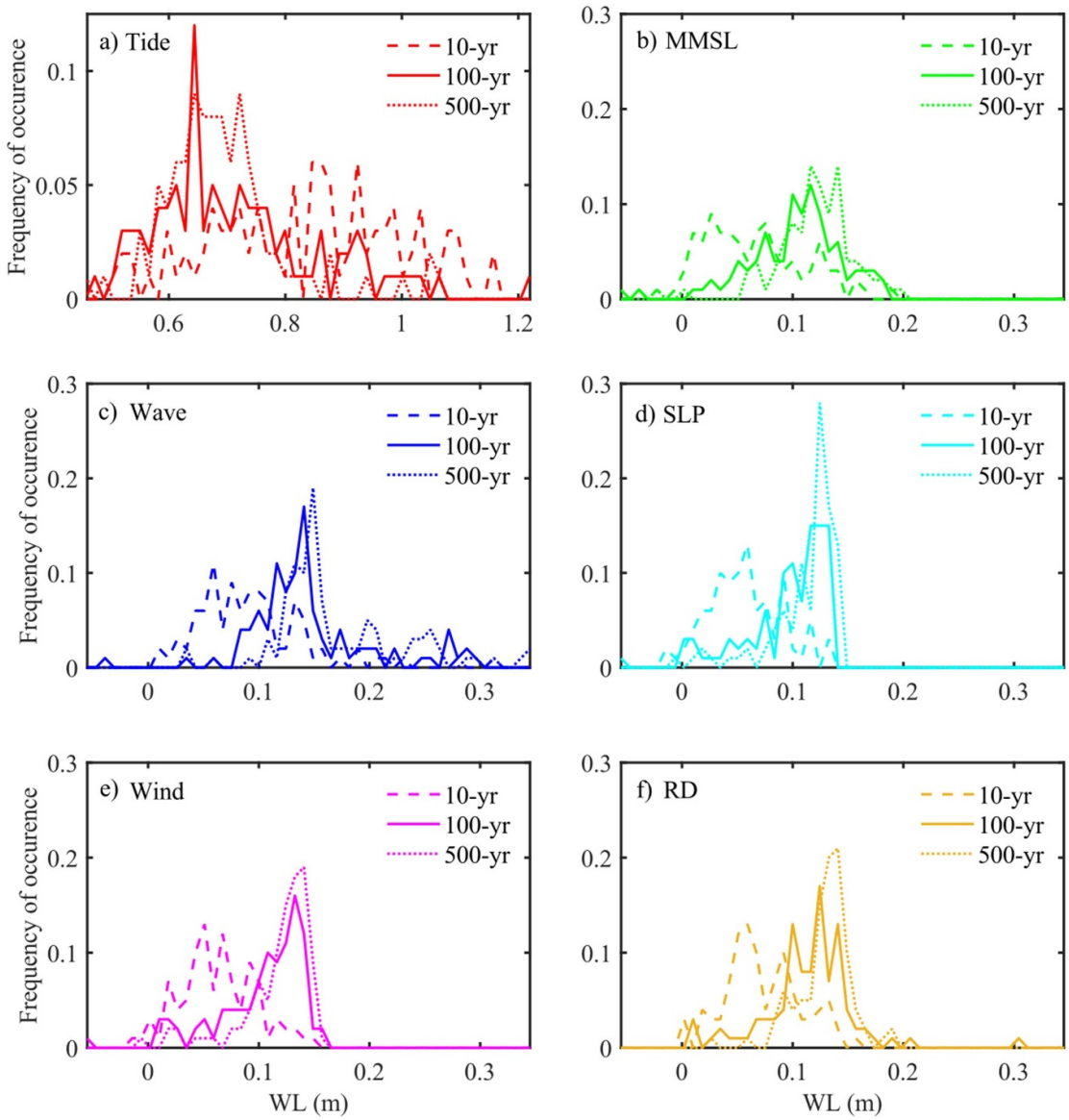

Figure 5. Probability distribution of WLs due to (a) tide, (b) MMSL, (c) wave, (d) SLP, (e) wind, and (f) RD at location 5 over the past 500 years.

Near the Delta mouth, locations 1 and 2 may experience more severe minor and moderate flooding with a higher frequency than location 3 due to higher extreme TWLs (e.g. in 500 yr period shown in figure 3(d)) but the same flood threshold. The overall higher flood frequency in South Bay confirms that this ends of SF Bay are more vulnerable to flooding [55]. Flood frequency analysis is useful to inform flood protection design, e.g. designing flood protection infrastructure based on appropriate return level events that account for different vulnerabilities of different locations [56, 57]. Note that moderate and major flooding occur with much less frequency than nuisance and minor flooding in the analysis; however, moderate and major flooding are expected to become more probable with climate change (e.g. effect of climate change on river floods and sea level rise) in the future [58].

The increase of the relative contributions of non-tidal drivers and the decrease of the contribution of tides with the increase of return period is caused by the larger magnitudes of non-tidal drivers (especially RDs near the Delta) when they experience rare events in longer return periods [11, 59]. This also explains why non-tidal drivers have a larger contribution to the 500 yr return level event than tides near the Delta. The comparison between the contributions of tides and the non-tidal drivers reconfirms that extreme compound flooding is not necessarily caused by situations in which all drivers are extreme [1, 24]. These extreme flood events identified by the proposed hybrid approach might not be fully captured using either a dynamical approach or a statistical approach alone that assumes that extreme flooding is induced by a particular forcing combination with extreme values of drivers [1, 11]. The tidal

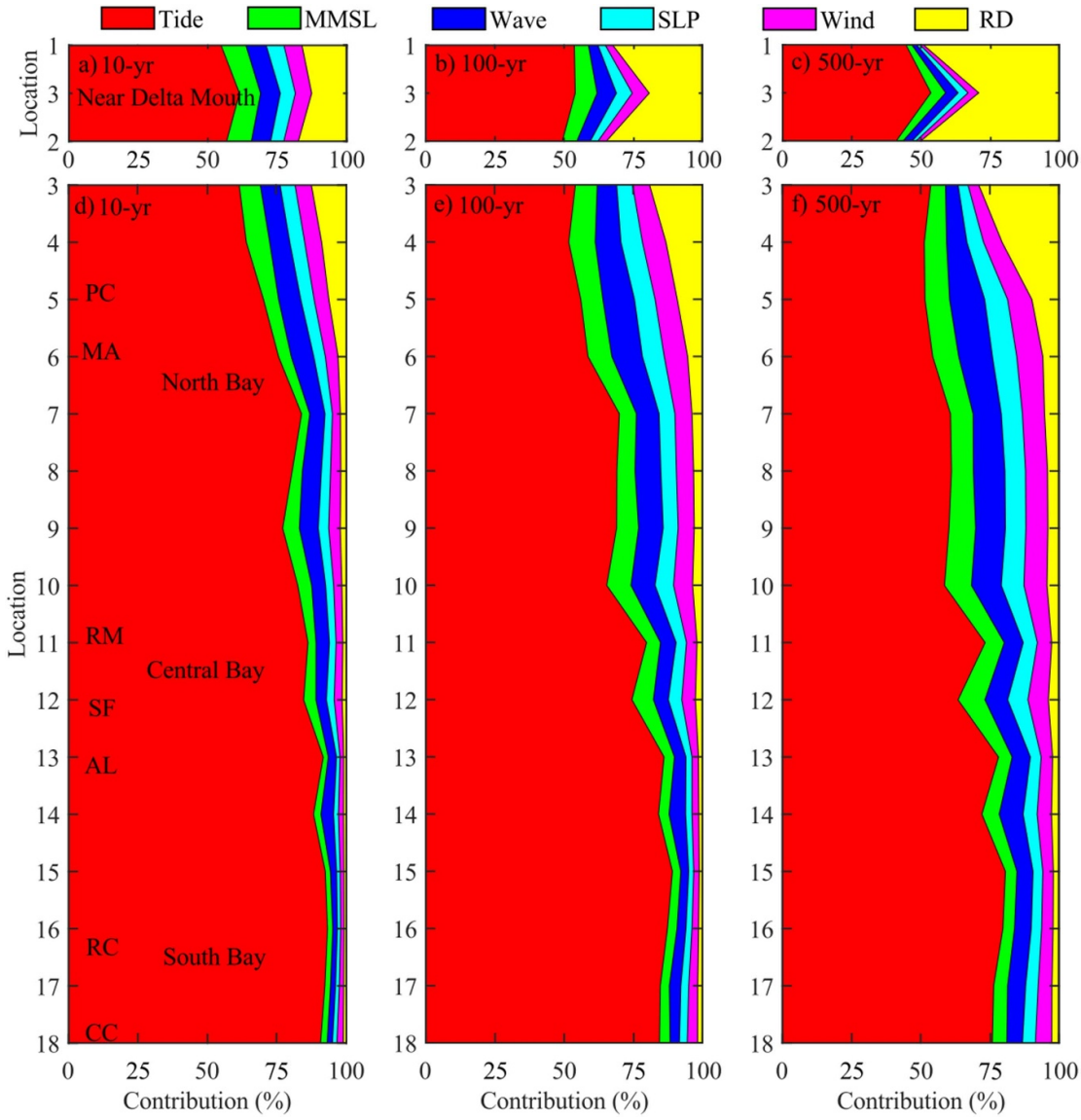

Figure 6. (a)–(c) and (d)–(f) Spatial distribution of the mean relative contributions of different drivers to extreme TWLs near the Delta mouth and along the centerline of SF Bay.

dominance to the extreme TWL has been observed on the U.S. West Coast [60]. The increase of tides' contribution towards the south was also found in [11] and the increase of non-tidal drivers towards the Delta is in line with the analysis in [11, 61]. The finding that the peak in the non-tidal relative contribution distributions are all around 0.1 m at location 5 (figure 5) indicates that each of the drivers examined here (MMSL, waves, SLP, wind, and RD) are important for understanding extreme TWLs in SF Bay. The overall similar relative contributions of the non-tidal drivers to the non-tidal water level signal at locations 8 through 18 further shows that all non-tidal drivers are of similar importance and none of them can be ignored when analyzing extreme TWLs in SF Bay. Note that RD has a significantly larger magnitude during the less frequent extreme events (hihger return level events), leading to a larger contribution to non-tidal water level signal than other non-tidal drivers near the Delta. This finding highlights the fact that RD has important consequences on compound flooding on the upstream boundary [62]. It further highlights the power of the proposed hybrid framework for compound coastal-fluvial flooding analysis by incorporating the weather generator and the hydrologic and reservoir system model to generate a large size of samples for RD that may contribute to extreme flood events. Assessing the relative contribution of each driver to extreme TWLs will help to better understand how the change in individual driver due to climate change might affect compound flooding, which is crucial to inform effective flood adaptation [63].

The hybrid statistical–dynamical framework presented here is developed by integrating different models/emulators through a modular design approach. These models/emulators are general and can be adapted to different coastal areas by updating

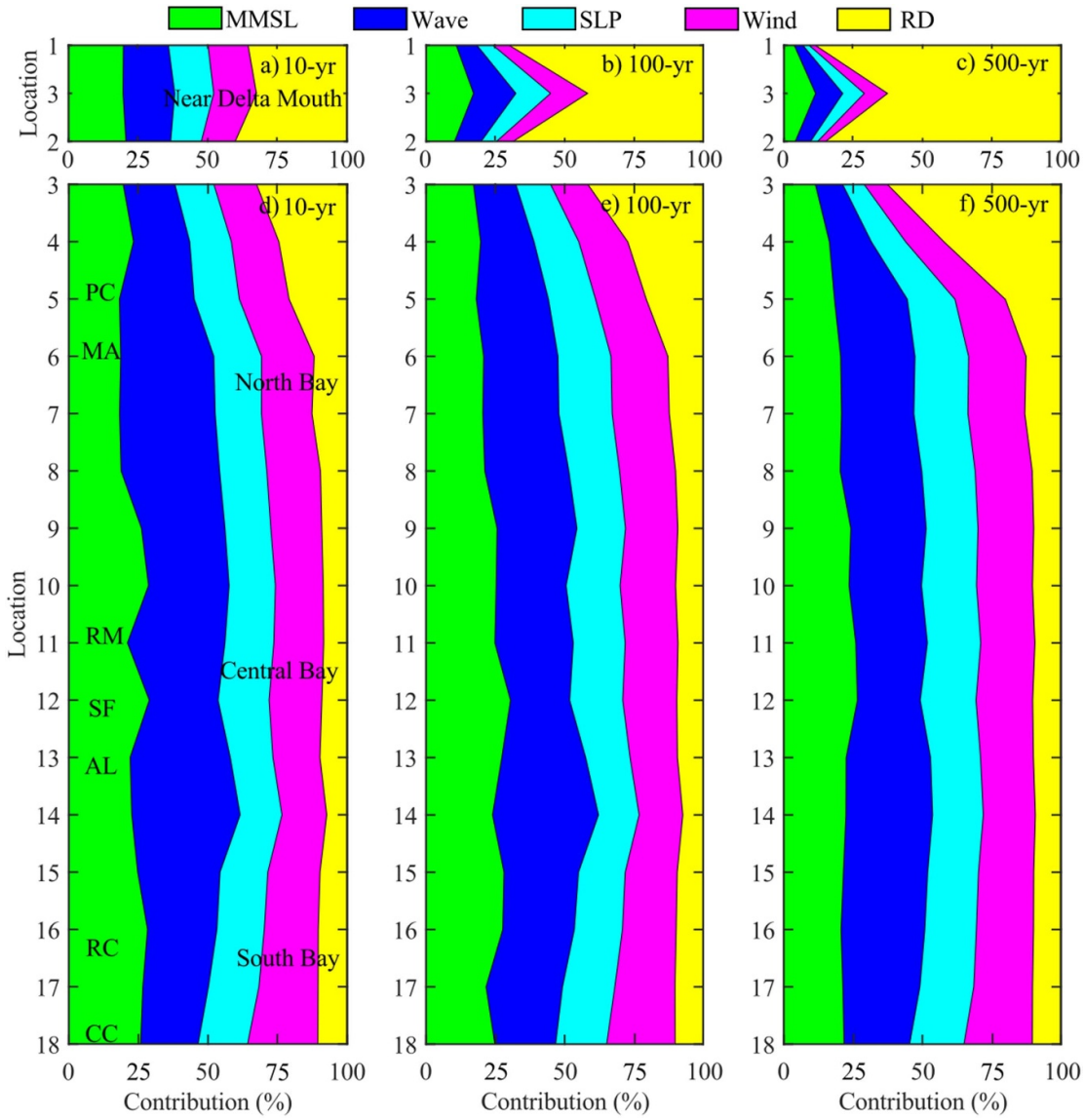

Figure 7. (a)–(c) and (d)–(f) Spatial distribution of the mean relative contributions of non-tidal drivers to the non-tidal water levels associated with extreme TWLs near the Delta mouth and along the centerline of SF Bay.

different settings accordingly. Therefore, the hybrid framework is generalizable and has broader applicability to other coastal regions for flooding analysis, especially coastal areas with compound coastal-fluvial flooding. For a new coastal region, the required data can first be fed into each model/emulator of the stochastic compound flooding driver generator to create synthetic time series of various drivers specific to the site over a period of interest. Note that the weather generator and the hydrologic and reservoir system model are not needed if no RD flows into the region. There is also no need to rerun the SST reconstruction model to inform TESLA to create the annual weather type (AWT) time series if the period of interest falls within the past 500 years since AWTs are assumed similar in this approach throughout the Pacific basin. Stochastic sampling of some representative samples (forcing combinations) is straightforward and is not dependent on the specific study site. A high-fidelity hydrodynamic model (e.g. D3D FM coupled flow-wave model) can be developed for new site with its geological characteristics (e.g. topography, bathymetry, bottom friction, and flood defense) and the boundary conditions defined by representative samples of multiple drivers to simulate the TWL at any location in the study area. Based on the input and output of the hydrodynamic model (representative forcing combinations and the resulting TWLs), surrogate models such as GPR models can be built for the location of interest to efficiently simulate the WLs due to a full range of forcing combinations. The surrogate models are statistical and their training is not dependent on the specific site. However, model validation needs to be performed using historical data (e.g. observed TWLs at NOAA gauges) at each new site. Finally, compound flooding analysis

can be conducted for new sites in terms of different flood metrics based on the large size of samples (e.g. TWL time series).

There are many interesting avenues for improving the proposed hybrid framework for compound coastal flooding analysis. On the one hand, the hybrid framework can be made more powerful for understanding the complex spatial-temporal dynamics of compound coastal flooding by improving the surrogate model. To avoid introducing additional complexity to the hybrid framework in this application, the GPR surrogate models are developed by assuming stationary covariances with low-dimensional (i.e. 25) inputs based on the steady-sate rather than synthetic time series of boundary conditions for the hydrodynamic model. The flooding analysis supported by this surrogate modeling approach may not capture the complex temporal and nonlinear dynamics of compound flooding in the Bay. This could be addressed by building the surrogate model using a nonstationary covariance function and synthetic time series of training data [64]. Based on the low-dimensional output of the GPR models (i.e. TWL), the compound flooding analysis is illustrated at 18 representative locations along the centerline of SF Bay rather than the entire Bay including floodplains. This flooding analysis may not capture the complex spatial dynamics of compound flooding in the Bay. Both issues with spatiotemporal dynamics of flooding could be addressed by developing a surrogate model that can enable spatial flood mapping in urban settings using more advanced machine learning techniques such as neural networks (e.g. combining recurrent neural network [15] and convolutional neural networks [65]) based on high-dimensional inputs (i.e. the synthetic time series of drivers) and outputs (i.e. spatial TWLs at all locations). In addition, advanced machine learning techniques such as transfer learning [66–68] may be employed to build a generalized surrogate model for accurate compound flooding prediction in a new location with similar characteristics rather than building additional surrogate models for different coastal regions. White-box machine learning models instead of black-box models may be built to capture the non-linear interaction between drivers more accurately and hence to spot the relative importance of drivers to extreme TWLs [63].

On the other hand, several aspects that may influence the uncertainty of results while are not the focus of this study could be incorporated to better understand the flooding analysis results. The generator of compound flooding drivers can be improved to account for the non-stationarity of forcing variables, which may influence flood return periods [69, 70]. The generator may also be improved to more rigorously propagate the cascading uncertainty in the modeling chain. The D3D FM model does not consider the evolution and/or change of the physical conditions including topography, bathymetry (dredging), land cover, hydrologic connectivity, anthropogenic activities, land subsidence, flood defenses, etc since the complexity (e.g. surrogate models with high dimensional inputs) is beyond the scope of the study. However, these temporally varying conditions can have a significant impact on the TWL variability. Ignoring spatial variability of winds and SLP for the dimensionality reduction in the development of surrogate model may also introduce uncertainty in TWL variability and compound flood dynamics. The bathymetry can also play an important role in the interaction of non-tidal drivers, especially during a long time period (e.g. 500 years). In addition, climate drivers such as Pacific decadal oscillation and North Pacific index (NP index) [71–73] may impact the local weather patterns (i.e. DWTs here) while they are not taken into account in the current version of TESLA. All the above factors that may impact the uncertainty of results can be incorporated into the hybrid framework in future work to better understand the compound coastal flooding.

6. Conclusion

By combining a stochastic generator for coastal flooding drivers, a physics-based hydrodynamic model, and machine learning-based surrogate models, this study develops a hybrid statistical-dynamical framework for spatiotemporal analysis of compound coastal flooding in an efficient and accurate manner. Using this hybrid approach, compound coastal flooding in SF Bay representative of the past 500 years was analyzed along the centerline of the Bay, with the focus on extreme TWLs, flood frequency, and the relative contributions of different drivers to extreme TWLs. Findings include that extreme TWLs have a smaller variability as return period increases and that nuisance and minor flooding occur with different return periods at most locations within the Bay. The relative contributions of different drivers display a smaller variability with the increase of return period. As shown by other studies, tides dominate the extreme TWLs, and the relative contribution of tides increases towards the south. The contribution of tides becomes smaller while the contribution of non-tidal drivers (MMSL, waves, SLP, wind, and RD) becomes larger as longer return periods are examined. RDs dominate the non-tidal water levels associated with extreme events near the Delta. For much of the rest of the Bay, the remaining drivers have roughly the same order of contribution to the non-tidal components of extreme TWLs. The hybrid statistical–dynamical framework presented here could be generalized to other complex estuaries for a comprehensive investigation of possible forcing combinations for extreme compound flooding events that might not be captured by physics-driven approaches or data-driven statistical approaches. The hybrid approach based spatiotemporal analysis of compound flooding can

provide important information about extreme design events, which would be useful for flood protection planing and design in different return periods of interest.

Data availability statement

A large dataset from a variety of sources have been used for this research. The data that support the findings of this study are available upon reasonable request from the authors.

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